

ARCHIT AGRAWAL

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AI Software Engineer with 4+ years building production agent systems, retrieval infrastructure, and full-stack products. Specializes in turning experimental AI workflows into reliable systems through evaluation, schema enforcement, idempotency, observability, and measurable performance.

EXPERIENCE

AI Software Engineer - EdPlus, Arizona State University

Tempe, AZ | Jun 2026–Present

- Architected and shipped a reusable survey-analysis platform of 10 agents and 65 registered tools; replaced bespoke pipelines with a declarative TypeScript architecture on Bedrock, AgentCore, Step Functions, and CDK.
- Prevented roughly 12% silent row loss through schema-constrained decoding, response correlation tokens, and citation verification; achieved 100% join integrity and zero label churn across repeat runs.
- Made newly uploaded surveys immediately queryable without dataset-specific configuration by building a natural-language semantic layer over DuckDB.
- Held query latency flat at 200× row volume using in-process DuckDB over Lance on S3.
- Redirected the team roadmap through a controlled evaluation showing model choice moved labeling F1 by 0.230 - roughly 33× the 0.007 movement from pipeline architecture.
- Eliminated a Distributed Map race that reported success while dropping data through partitioned writes and fan-in, verified across 258 runs.
- Delivered the platform through a 14-page Nuxt 4 application and 10-module NestJS API compiled against one shared TypeScript contract.

Founding AI/ML Engineer - MyStage Music Inc.

Remote | Jul 2025–Present

- As founding AI/ML engineer, cut service dependencies 64% (14 to 5) by re-architecting a 14-Cloud-Function pipeline as one checkpointed LangGraph worker on Cloud Run.
- Indexed 70,000+ records daily across 650+ venues through a Playwright service with proxy rotation and adaptive backoff.
- Improved downstream search accuracy 25% by resolving and deduplicating entities with Gemini before Firestore ingestion.
- Served search at sub-50 ms latency through FastAPI and Algolia with real-time Firestore synchronization.
- Made concurrent retries safe with atomic Firestore claims and deterministic task IDs; added human-in-the-loop pause and resume.
- Made failures diagnosable and changes safe with Logfire distributed tracing and 200+ pytest tests.

Software Engineer - EdPlus, Arizona State University

Tempe, AZ | Sep 2023–May 2025

- Led the end-to-end development of a multi-tenant RAG assistant used by 1,000+ faculty authoring courses for 60,000+ students.
- Made transcript analysis 16× faster, reducing it from four hours to 15 minutes with a Neo4j knowledge graph and schema-constrained, LLM-generated Cypher.
- Reduced college onboarding from a deployment to a configuration change while enforcing tenant isolation at retrieval rather than prompt level.
- Improved exact-token retrieval for queries such as course codes with hybrid vector and keyword search; evaluated groundedness, relevance, and coherence.

Data Research Aide - Knowledge Exchange for Resilience, ASU

Tempe, AZ | Jun 2024–Aug 2024

- Reduced p95 retrieval latency 60% at 95% recall with hybrid FAISS and Neo4j retrieval over Postgres/pgvector.
- Reduced API latency nearly 90%, from 198 ms to 20 ms, through Redis and SQL optimization, verified under Locust load.
- Delivered 20+ .NET 8 REST APIs for an NSF-funded platform and an idempotent ETL pipeline ingesting 10,000+ faculty profiles daily.

Software Engineer - Zeus Learning

Mumbai, India | Jan 2022–Jul 2023

- Reduced resource usage 35% and infrastructure cost 20% by migrating a .NET monolith to Kubernetes microservices through a strangler pattern.
- Reduced deployment time 70% and incidents 40% with SonarQube merge gates and containerized delivery.
- Shipped a Redis-backed demand-prediction service across 300+ enterprise sites, including Goldman Sachs and Merck; reported occupancy increased 30%.

Product Intern - EAT.FIT

Bengaluru, India | Sep 2021–Dec 2021

- Reduced driver-location serving cost 45% through distance-aware polling and client interpolation.
- Built a backpressure-aware WebSocket notification service with idempotent redelivery.

EDUCATION

M.S., Computer Science, Arizona State University - GPA: 4.0

Aug 2023–May 2025

B.Tech., Information and Communication Technology, DA-IICT

Aug 2018–May 2022

SELECTED PROJECTS

MCP GitHub PR Review Agent

- Built a webhook-triggered MCP service that combines repository state, ticket metadata, and CI context to automate review decisions.
- Designed the tool boundary so review logic receives typed, current repository evidence rather than relying on prompt-only context.
- Separated evidence collection from review policy so repositories can change review criteria without rebuilding the integration layer.

SRP Electric MCP Server

- Exposed an undocumented utility portal as validated TypeScript tools for secure energy-data analysis by AI agents.
- Reverse-engineered the authentication flow and translated unstable portal behavior into typed tool contracts with explicit validation.
- Normalized legacy responses behind MCP schemas so downstream agents operate on predictable data rather than portal-specific markup.

Video Generation Physics Evaluation - ASU Research

- Built an evaluation harness that scores physical plausibility separately from semantic fidelity instead of hiding both behind one aggregate score.
- Designed controlled prompt comparisons and found counterfactual prompts degraded faster than descriptive prompts at matched length.
- Automated frame extraction and artifact capture so every score remains traceable to the generated evidence that produced it.

PrismSplit - Item-Level Expense Sharing

- Built a React Native and TypeScript product that scans receipts, assigns individual items, and calculates multi-party balances using integer-cent arithmetic.
- Kept bill creation and settlement transactional through Postgres RPCs; added idempotency keys, offline persistence, real-time synchronization, and Maestro release tests.
- Implemented debt-edge reduction to minimize settlement transactions and a shared web target over the same stores and services.

AiJockey - AI DJ Pipeline

- Built an end-to-end FastAPI pipeline for stem separation, musical analysis, LLM-directed transition planning, DSP rendering, and mastering.
- Evaluated rendered variants with audio-quality critics and trained a MERT reward head to predict aesthetic quality during selection.
- Ran PyTorch, Demucs, VampNet, MERT, and Audiobox in a ROCm container on an MI300X GPU; checkpointed artifacts for rapid recovery.

ADDITIONAL RESEARCH

Student Researcher - Arizona State University

Tempe, AZ | Aug 2024–May 2025

- Designed minimal-pair prompts and a failure taxonomy spanning object permanence, event ordering, counterfactual conditions, and rigid-body motion.
- Measured inter-rater agreement before aggregating results and held seeds, resolution, frame count, and sampling settings fixed across model comparisons.

Computer Vision Researcher - DA-IICT

Gandhinagar, India | May–Aug 2021

- Worked across radar/lidar capture, timestamp alignment, point-cloud processing, sensor calibration, and HD-map localization for autonomous-driving research.
- Built a replay harness over recorded sensor logs so perception changes could be evaluated against identical drives and attributable failure cases.

SKILLS

Languages: Python, TypeScript, C#, SQL, JavaScript

AI & Data: LangGraph, MCP, Bedrock/AgentCore, Vertex AI, OpenAI, RAG, DuckDB, PostgreSQL/pgvector, Neo4j, FAISS, Lance, Algolia, Redis, Firestore

Backend & Cloud: FastAPI, NestJS, .NET 8, React, Nuxt 4, AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions

Quality: pytest, Vitest, Locust, Logfire, distributed tracing